

11-Week Development Plan for Simple Restaurant Management Web App

Project Overview

This is a basic restaurant management web app for a manager. The manager can:

- View food items by category
- Add, edit, delete food items
- Increment or decrement quantity of food items
- Generate a bill from selected items

Technologies used: Node.js, Express, MySQL (Backend) and ReactJS (Frontend)

11-Week Development Plan

Week 1 – Project Setup + HTML Form Design

- Tasks:
 - Create basic HTML forms (Food entry: Name, Price, Category, Description)
 - Style using CSS or Bootstrap
- Deliverables:
 - foodForm.html
 - style.css

Week 2 – JavaScript Form Functionality

- Tasks:
 - Validate input fields using JavaScript
 - Store data in array and show in HTML table
- Deliverables:
 - app.js with JS-only CRUD

Week 3 – Backend with Node.js + ExpressJS

- Tasks:
 - Setup NodeJS, ExpressJS project
 - Create endpoints: /api/foods GET, POST
- Deliverables:
 - Running Express server with JSON routes

Week 4 – MySQL Integration

- Tasks:
 - Setup MySQL and restaurant_db
 - Create foods and categories tables and connect MySQL to Express
- Deliverables:
 - db.js, SQL schema script, working API

Week 5 – Full CRUD API for Food

- Tasks:
 - Implement RESTful endpoints: GET, POST, PUT, DELETE
- Deliverables:
 - Tested CRUD API for foods

Week 6 – Build React Frontend (UI Only)

- Tasks:
 - Create React app with Vite/CRA
 - Create pages: FoodList, AddFood, EditFood using static data
- Deliverables:
 - React pages and routes created

Week 7 – Integrate Backend API with React

- Tasks:
 - Use Axios/Fetch to connect to backend API
 - Perform Add and View operations
- Deliverables:
 - React + API working integration

Week 8 – Food Category & Quantity Buttons

- Tasks:
 - Group food items by category
 - Implement + and - buttons for quantity control
- Deliverables:
 - Working quantity UI and React state update

Week 9 – Generate Bill Functionality

- Tasks:
 - Generate bill from selected items with quantities and total price
 - Optionally save to database
- Deliverables:
 - Bill component with calculation

Week 10 – Styling & Form Validations

- Tasks:
 - Use Bootstrap or custom CSS for styling
 - Add form validations
- Deliverables:
 - Styled UI with validations

Week 11 – Final Testing & Deployment

- Tasks:
 - Host backend on Render or Cyclic
 - Host frontend on Vercel or Netlify
 - Prepare GitHub repo, ERD, documentation
- Deliverables:
 - Live URLs, GitHub repo, documentation

Extra Add-Ons (Optional)

- PDF Bill generation
- Role-based login (manager only)
- Category-wise filters with dropdown
- Real-time update with Socket.io